

SAN FRANCISCO HARBOR, CA, USA

WMO: 994016

Lat: 37.806N

Lon: 122.466W

Elev: 7

StdP: 14.69

Time Zone: -8.00 (NAP)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) 41.6	(c) 43.2	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 24.2	(k) 53.7	(l) 20.0	(m) 52.8	(n) 6.6	(o) 200	(p) 0.531

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 9	(b) 7.2	(c) 72.0	(d) N/A	(e) 68.3	(f) N/A	(g) 65.8	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 7.3	(p) 250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 20.6	(b) 18.2	(c) 16.3	(e) 39.1	(f) 81.5	(g) 2.0	(h) 3.6	(i) 37.6	(j) 84.1	(k) 36.4	(l) 86.2	(m) 35.3	(n) 88.3	(o) 33.8	(p) 90.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	55.1	51.4	51.8	52.6	53.2	54.5	56.2	57.3	58.4	59.3	59.1	55.3	51.2
	DBStd	4.58	3.78	4.18	3.47	3.20	3.14	2.86	2.63	2.95	3.73	4.33	3.99	3.98
	HDD50	107	29	26	11	4	0	0	0	0	0	0	4	33
	HDD65	3654	420	370	385	354	327	265	240	205	179	191	290	428
	CDD50	1954	74	76	92	102	140	187	225	261	278	284	165	70
	CDD65	23	0	0	0	1	2	2	0	1	7	9	1	0
	CDH74	70	0	0	0	6	6	9	0	8	20	21	0	0
	CDH80	7	0	0	0	0	1	1	0	2	1	2	0	0
Wind	WSAvg	7.5	7.1	6.6	7.0	7.8	8.4	9.3	9.0	8.3	7.4	6.2	6.0	6.7
Precipitation	PrecAvg	20.5	4.5	3.7	2.9	1.3	0.4	0.1	0.0	0.1	0.2	1.1	2.8	3.5
	PrecMax	38.3	11.3	13.6	8.7	5.3	2.4	0.9	0.4	0.7	1.3	7.3	8.0	10.7
	PrecMin	9.8	0.2	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	6.9	3.0	3.0	2.1	1.3	0.6	0.2	0.1	0.2	0.3	1.4	2.2	2.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.3	66.2	66.9	72.3	72.3	74.6	66.4	70.9	77.1	76.9	70.6	63.6
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	60.4	62.4	61.7	63.4	64.1	64.6	64.3	66.2	69.3	71.0	66.2	60.7
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	58.5	59.8	59.3	59.3	60.5	61.6	62.8	64.2	66.0	67.4	62.9	58.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	56.8	57.6	57.5	57.4	58.7	60.1	61.3	62.5	64.0	64.4	60.6	57.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	7.8	8.3	8.3	7.9	6.9	6.7	5.3	5.5	7.2	8.7	8.2	7.5
		MCDBR	10.2	11.3	10.9	11.7	11.2	9.8	6.7	7.7	12.9	14.6	11.7	8.6
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub		0.323	0.329	0.340	0.357	0.363	0.357	0.367	0.344	0.345	0.337	0.322	0.324
	taud		2.564	2.545	2.506	2.423	2.436	2.457	2.434	2.499	2.482	2.548	2.569	2.567
	Ebn at Noon		275	287	293	292	291	292	288	293	287	280	272	266
	Edh at Noon		24	27	31	36	36	35	36	33	31	27	23	23
All-Sky Solar Radiation	RadAvg		756	1050	1482	1944	2225	2423	2276	2023	1726	1276	887	658
	RadStd		105	129	133	152	148	107	82	95	68	84	59	97

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (44 neighbors)		N/A	N/A	-2.14	N/A	N/A	N/A	N/A	-198	+243	N/A

Nomenclature: See separate page